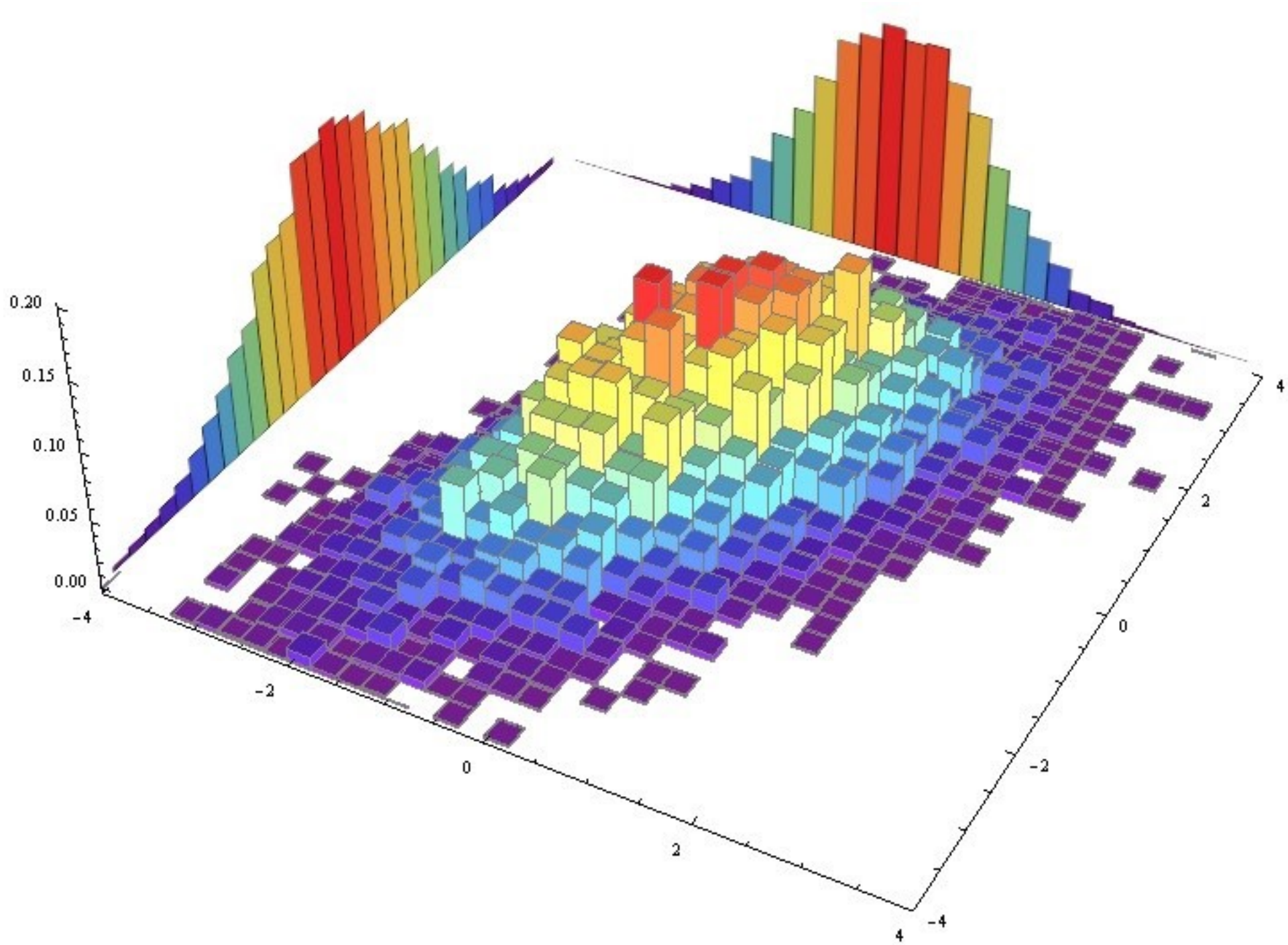
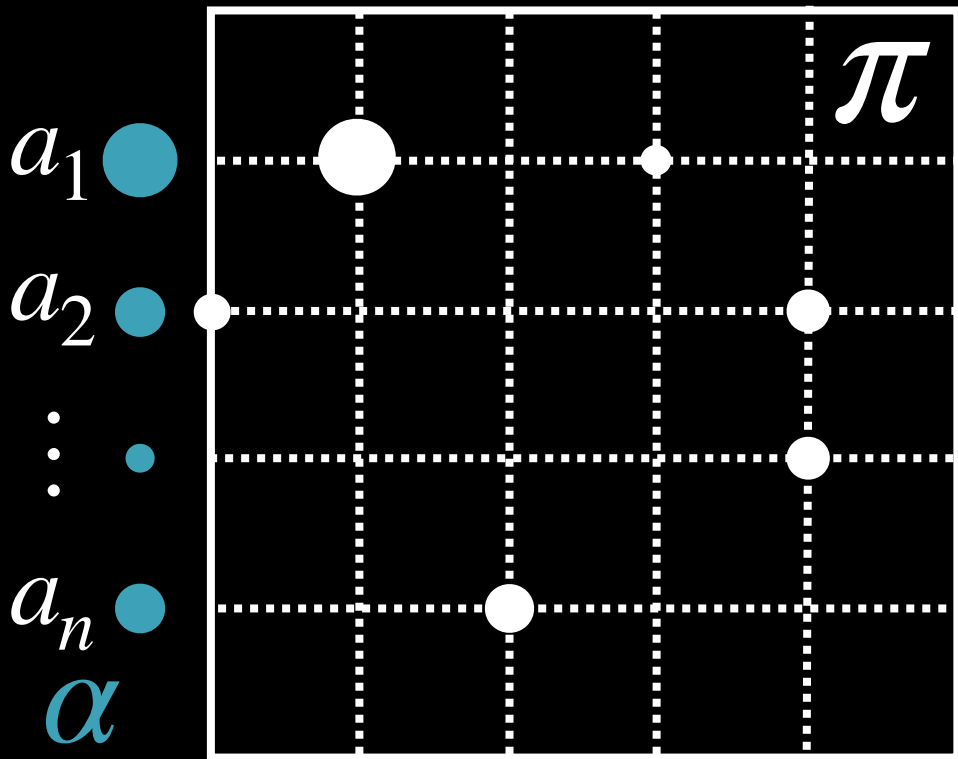




b_1 b_2 $\dots$ b_m β

$$\alpha = \sum_1^n a_i \delta_{x_i}, \quad a \in \Delta_n,$$

$$\beta = \sum_1^m \mathbf{b}_j \delta_{y_j}, \mathbf{b} \in \Delta_m$$

Pop-Quiz: Double Sums

Prob. measures:

$$\alpha = \sum_1^n \mathbf{a}_i \delta_{x_i}, \mathbf{a} \in \Delta_n,$$

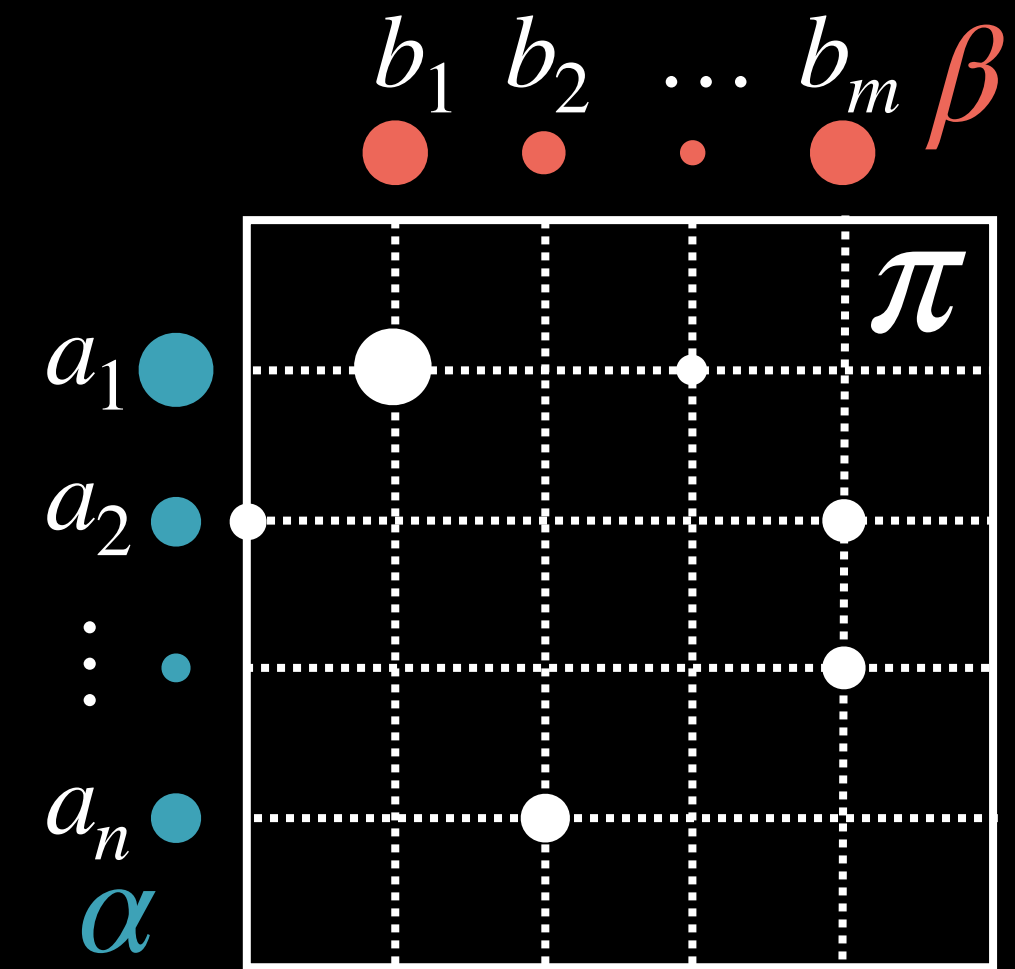
$$\beta = \sum_1^m \mathbf{b}_j \delta_{y_j}, \mathbf{b} \in \Delta_m$$

Let $\pi \in \mathbb{R}_+^{n \times m}$ be a valid coupling between α, β . Then

$$\sum_{i=1}^n \sum_{j=1}^m \pi_{i,j} = ?$$

Answer: $\sum_{i=1}^n \sum_{j=1}^m \pi_{i,j} = 1.$

Proof: $\sum_{i=1}^n \sum_{j=1}^m \pi_{i,j} = \mathbf{1}_n^\top \pi \mathbf{1}_m = \mathbf{1}_n^\top \mathbf{a} = 1$



$$\sum_{j=1}^m \pi_{ij} = a_i$$

$$\sum_{i=1}^n \pi_{ij} = b_j$$

Examples of Couplings

Permutations

$$\sigma \in S_n$$